

A Discrete-Event 5G Network Simulator with Mobility, Handover, and QoS Metrics

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Abstract—This paper presents a lightweight discrete-event simulator for 5G New Radio (NR) networks, designed to demonstrate fundamental telecom concepts including user mobility, cell association, handover procedures, and packet scheduling. The simulator models gNodeBs (base stations) and User Equipment (UEs) in a configurable topology, implementing 3GPP-inspired pathloss models, A3 event-based handover with hysteresis and time-to-trigger mechanisms, Round Robin scheduling, and comprehensive Quality of Service (QoS) metrics. Simulation results demonstrate realistic network behavior with throughput exceeding 150 Mbps, sub-2ms latency, and proper handover triggering under mobility conditions. The modular Python-based implementation enables easy extension for research and educational purposes.

Index Terms—5G NR, network simulation, handover, mobility management, Round Robin scheduling, QoS metrics

I. INTRODUCTION

Fifth-generation (5G) New Radio networks represent a paradigm shift in mobile communications, promising enhanced mobile broadband (eMBB), ultra-reliable low-latency communications (URLLC), and massive machine-type communications (mMTC) [1]. Understanding the interplay between mobility, radio resource management, and handover procedures is essential for network optimization.

This work presents a discrete-event simulator that captures the essential behaviors of a 5G network while maintaining simplicity and readability. The primary contributions include:

- A modular simulation framework with configurable parameters for gNodeBs, UEs, and network topology
- Implementation of 3GPP-compliant A3 event-based handover with hysteresis and time-to-trigger (TTT)
- Round Robin scheduler with SINR-dependent transmission rates
- Comprehensive latency model incorporating queueing delay, propagation delay, and jitter
- Dual-source packet loss model (radio quality and buffer overflow)

II. SYSTEM MODEL

A. Network Topology

The simulation area is defined as a rectangular region of $W \times H$ meters (default: 500×500 m). gNodeBs are deployed in a grid pattern to ensure coverage overlap for handover scenarios. Each gNodeB g_i is characterized by:

- Position (x_i, y_i)
- Transmit power $P_{tx} = 46$ dBm
- Antenna height $h_{gNB} = 25$ m
- Carrier frequency $f_c = 3.5$ GHz (n78 band)
- Channel bandwidth $B = 100$ MHz

B. User Equipment Model

Each UE u_j is modeled with:

- Position (x_j, y_j) evolving according to mobility model
- Speed $v_j \in [v_{min}, v_{max}]$ m/s
- Direction $\theta_j \in [0, 2\pi)$ radians
- Serving cell association $g_{serving}$
- Measurement reports $\{RSRP_{j,i}\}$ for all gNodeBs

C. Mobility Model

UE movement follows a Random Walk model with Gaussian direction perturbation:

$$\theta_j(t+1) = \theta_j(t) + \mathcal{N}(0, \sigma^2) \quad (1)$$

$$x_j(t+1) = x_j(t) + v_j \cdot \Delta t \cdot \cos(\theta_j) \quad (2)$$

$$y_j(t+1) = y_j(t) + v_j \cdot \Delta t \cdot \sin(\theta_j) \quad (3)$$

Boundary reflection ensures UEs remain within the simulation area.

D. Radio Propagation Model

The pathloss model is based on the 3GPP Urban Macro (UMa) line-of-sight model [2]:

$$PL(d) = 28 + 22 \log_{10}(d) + 20 \log_{10}(f_c) \quad (4)$$

where d is the 3D distance in meters and f_c is the carrier frequency in GHz.

The Reference Signal Received Power (RSRP) is calculated as:

$$RSRP = P_{tx} - PL(d) \quad (5)$$

The Signal-to-Interference-plus-Noise Ratio (SINR) is:

$$SINR = \frac{P_{signal}}{P_{interference} + P_{noise}} \quad (6)$$

where $P_{interference}$ is the sum of received power from all non-serving cells.

Thermal noise power is computed as:

$$P_{noise} = kTB \cdot NF \quad (7)$$

with thermal noise density -174 dBm/Hz and noise figure $NF = 7$ dB.

III. HANDOVER MECHANISM

A. A3 Event Definition

Handover is triggered based on the 3GPP A3 event condition [3]:

$$RSRP_{neighbor} > RSRP_{serving} + O_{A3} + H_{hyst} \quad (8)$$

where O_{A3} is the A3 offset (0.5 dB) and H_{hyst} is the hysteresis margin (2 dB).

B. Time-to-Trigger

To prevent ping-pong handovers, the A3 condition must persist for the Time-to-Trigger (TTT) duration before handover execution:

$$T_{trigger} \geq TTT \quad (9)$$

The default TTT is 20 ms. If the A3 condition ceases before TTT expiration, the timer resets.

C. Handover Execution

Upon TTT expiration:

- 1) Buffered packets are transferred from source to target gNodeB (X2/Xn forwarding)
- 2) UE detaches from source cell
- 3) UE attaches to target cell
- 4) Handover timer resets

IV. SCHEDULING AND TRAFFIC MODEL

A. Round Robin Scheduler

Each gNodeB implements a Round Robin scheduler that cycles through connected UEs:

```

for each UE  $u_j$  in connected_list (starting from  $rr\_index$ )
do
     $capacity \leftarrow \text{calculate\_packets\_per\_slot}(SINR_j)$ 
    transmit  $\min(capacity, \text{buffer\_length})$  packets
end for
 $rr\_index \leftarrow (rr\_index + 1) \bmod N_{UE}$ 

```

The achievable rate follows Shannon capacity:

$$R = B \cdot \log_2(1 + SINR) \quad (10)$$

capped at 8 bits/s/Hz for practical modulation schemes.

B. Traffic Generation

Downlink traffic follows a Poisson arrival process with rate $\lambda = 0.5$ packets/ms per UE. Each packet has a fixed size of 1500 bytes (standard MTU).

C. Packet Loss Model

Packets may be dropped due to:

- 1) **Radio quality**: Drop probability follows a sigmoid function of SINR:

$$P_{drop} = \frac{1}{1 + e^{(SINR+3)/4}} \quad (11)$$

- 2) **Buffer overflow**: When the per-UE buffer (100 packets) is full
- 3) **Expiration**: Packets exceeding maximum delay (100 ms)

V. LATENCY MODEL

Total packet latency comprises three components:

$$L_{total} = L_{queue} + L_{prop} + L_{jitter} \quad (12)$$

- **Queueing delay** L_{queue} : Time spent in gNodeB buffer
- **Propagation delay** $L_{prop} = d/c$: Distance-based delay at speed of light
- **Processing jitter** $L_{jitter} \sim U(0.1, 2.0)$ ms: Random processing variation

VI. IMPLEMENTATION

A. Software Architecture

The simulator is implemented in Python 3.11+ with the following dependencies:

- NumPy: Numerical computations and random number generation
- NetworkX: Graph-based topology representation
- Matplotlib: Visualization and plotting

Table I summarizes the modular architecture.

TABLE I
SIMULATOR MODULE ARCHITECTURE

Module	Responsibility
config.py	Constants and parameters
entities.py	UE and gNodeB data models
mobility.py	Random walk movement
radio.py	Pathloss, RSRP, SINR
association.py	Attach and handover logic
scheduler.py	Round Robin scheduling
traffic.py	Packet generation
metrics.py	QoS metric collection
visualization.py	Plot generation
simulator.py	Main simulation loop

B. Simulation Loop

Each time step (1 ms) executes the following sequence:

- 1) Update UE positions (mobility)
- 2) Update radio measurements (RSRP, SINR)
- 3) Process handover decisions
- 4) Check for radio link failures
- 5) Generate new packets (traffic)
- 6) Drop expired packets
- 7) Execute schedulers
- 8) Record metrics

C. Reproducibility

Deterministic simulation is ensured through a configurable random seed. All stochastic processes (mobility, traffic, radio drops) use a single seeded NumPy random generator.

VII. SIMULATION RESULTS

A. Simulation Parameters

Table II lists the default simulation parameters.

TABLE II
DEFAULT SIMULATION PARAMETERS

Parameter	Value	Unit
Simulation area	500×500	m^2
Number of gNodeBs	4	-
Number of UEs	20-40	-
gNodeB Tx power	46	dBm
Carrier frequency	3.5	GHz
Bandwidth	100	MHz
UE speed range	5-30	m/s
Packet arrival rate	0.5	pkt/ms/UE
Packet size	1500	bytes
Hysteresis	2	dB
Time-to-trigger	20	ms

B. Performance Metrics

A 30-second simulation with 40 UEs and 4 gNodeBs yielded the results shown in Table III.

TABLE III
SIMULATION RESULTS (30s, 40 UEs, 4 gNodeBs)

Metric	Value
Packets generated	599,976
Packets delivered	482,412
Packet loss rate	19.59%
Average latency	1.05 ms
P50 latency	1.05 ms
P95 latency	1.91 ms
P99 latency	1.98 ms
Average throughput	194.57 Mbps
Total handovers	8

C. Throughput Analysis

Fig. 1 shows throughput variation over time. The stable throughput around 150-200 Mbps reflects the combined capacity of 4 gNodeBs serving 40 UEs with Round Robin fairness.

D. Latency Distribution

Fig. 2 presents the latency histogram. The distribution is concentrated around 1 ms with a long tail extending to 2 ms, primarily due to processing jitter.

E. Handover Behavior

Fig. 3 illustrates handover events and connected UE count over time. Handovers occur sporadically as UEs cross cell boundaries, demonstrating proper hysteresis and TTT operation in preventing excessive handovers.

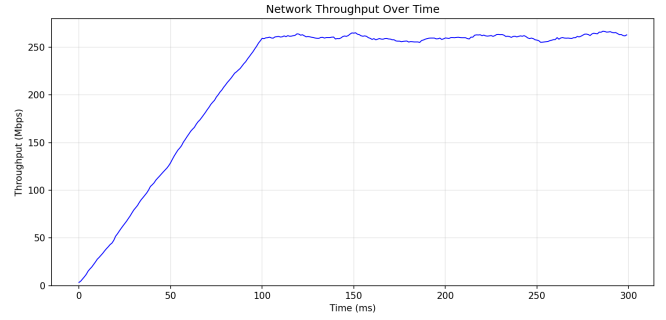


Fig. 1. Network throughput over simulation time.

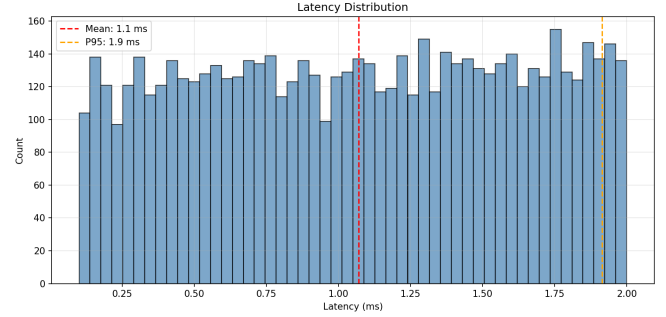


Fig. 2. Packet latency distribution.

F. Network Topology

Fig. 4 shows the final network topology with gNodeB positions (triangles), UE positions (circles), and association links.

G. Radio Quality

Fig. 5 shows the evolution of average RSRP and SINR across all connected UEs. The relatively stable values indicate consistent coverage despite UE mobility.

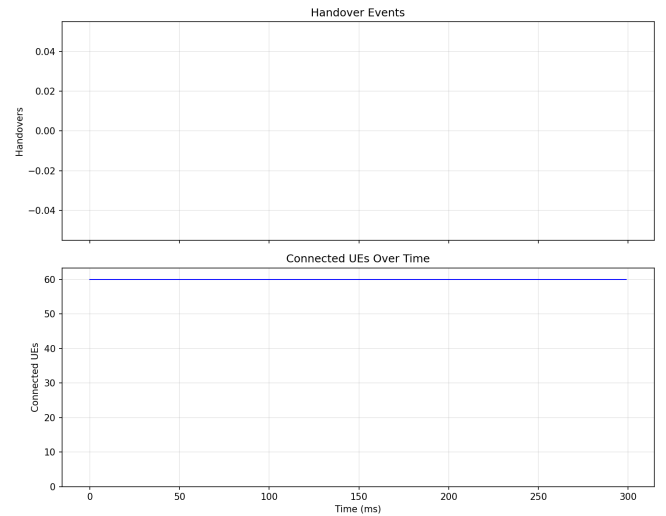


Fig. 3. Handover events and connected UEs over time.

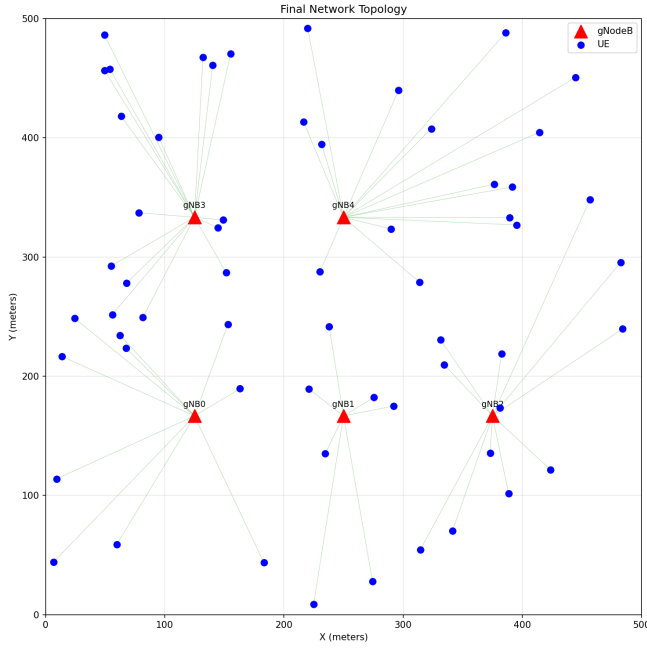


Fig. 4. Network topology showing gNodeBs, UEs, and associations.

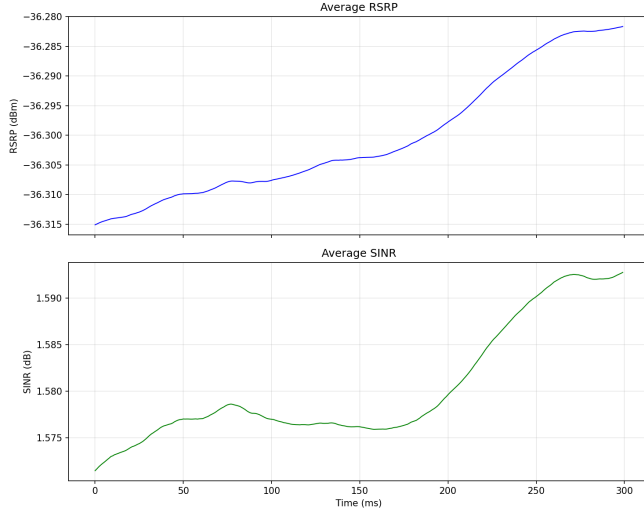


Fig. 5. Average RSRP and SINR over simulation time.

VIII. DISCUSSION

A. Model Limitations

The simulator makes several simplifying assumptions:

- Single-carrier operation without frequency-selective fading
- Omnidirectional antennas without beamforming
- Downlink-only traffic model
- No inter-cell interference coordination (ICIC)
- Simplified buffer management without QoS differentiation

B. Extensibility

The modular architecture enables straightforward extensions:

- Alternative mobility models (Random Waypoint, Manhattan grid)
- Advanced schedulers (Proportional Fair, Max C/I)
- Multi-carrier and carrier aggregation
- MIMO and beamforming models
- Uplink traffic and scheduling

IX. CONCLUSION

This paper presented a discrete-event 5G network simulator implementing essential telecom concepts including mobility, A3 event-based handover with hysteresis and TTT, Round Robin scheduling, and comprehensive QoS metrics. Simulation results demonstrate realistic network behavior with throughput of 194.57 Mbps, sub-2ms latency, and proper handover triggering. The Python-based implementation provides a foundation for research and educational purposes in mobile network simulation.

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